Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Early-career software engineer with 2+ years building **Python/Java backends**, JavaScript/React interfaces, data platforms, and AI applications; delivered healthcare, financial, and analytics systems handling 2M+ records with strong OOP, algorithms, SQL, and cloud fundamentals.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant data platform** using Python, FastAPI, Azure SQL, and Blob Storage for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through schema and indexing optimization.
- Restructured **fault-tolerant AI orchestration** with background workers, explicit job states, and failure isolation, processing 1,000+ concurrent jobs while improving throughput 3x and preventing cascading task failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **full-stack React and Flask workflows** for uploads, grading, history, and reporting, supporting 150+ users and 10,000+ daily requests while translating product needs into maintainable frontend and backend features.
- Integrated **multimodal AI and OCR providers** through switchable APIs, concurrent processing, usage logging, and fallback handling, reducing feedback turnaround 60% while improving reliability across production educational workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python data structures and evaluation pipelines** processing 100K+ trajectory records, integrating spatial-temporal features and reproducible simulation tests that improved long-horizon algorithm performance 18% across benchmark scenarios.
- Designed a **transformer-based planning architecture** combining map geometry and temporal context through multi-head attention, analyzing model behavior and failure patterns to improve closed-loop stability 15% across simulation experiments.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **object-oriented API and ETL components** using Singleton and Factory patterns, AWS Glue, S3, SQL, and caching, improving response times 30%, pipeline efficiency 68%, and maintainability across distributed workflows.
- Collaborated with **Microsoft and GitHub stakeholders** through cross-functional Agile sprints, maintaining 20+ unit/integration suites and Jenkins CI/CD across Windows, Linux, and macOS while reducing technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **React, Node.js, and Azure SQL features** across eight microservices for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs and WebSockets across locations.
- Automated **unit, integration, and end-to-end testing**, Docker packaging, and Jenkins deployments with staged approvals, cutting release time 50% while strengthening code quality across fast-paced iterative development cycles.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Core CS: Data structures, algorithms, OOP, design patterns, distributed systems, concurrency, system design
Full Stack: FastAPI, Flask, Node.js, Express.js, React.js, Next.js, REST APIs, WebSockets
Data & AI: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, ETL, LLM APIs, tool integrations, PyTorch
Delivery: Git, GitHub, Agile/Scrum, PyTest, Jest, Docker, Kubernetes, AWS, Azure, Jenkins, CI/CD

Projects

HMIS Live | *TypeScript, MERN, Healthcare APIs, Sentry*
- Built a **healthcare interoperability platform** with consent-based patient-history access, TypeScript dashboards, API integrations, appointment workflows, notifications, and monitoring, reducing record retrieval time 2x and administrative workload 30%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered data platform** supporting 10+ SQL and NoSQL systems, accelerating query authoring, schema exploration, and data management through a cross-platform Java and TypeScript codebase with Docker workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **financial data platform** supporting 50,000+ users and 1M+ transactions, combining secure bank integrations, real-time synchronization, advanced filtering, MongoDB access, responsive interfaces, and Kubernetes-based CI/CD automation.